

GROUP PROJECT REPORT

Project Title: __S-shape explorer Assistance____

Team Information

Name	StudentId/Email	Role/Responsibility
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 Person		
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Abstract

Write a short summary of the whole project in 150–250 words.

Checklist:

- [] State the problem.
- [] Mention the dataset.
- [] Mention the AI/ML method used.
- [] Summarize the main result.
- [] State the final output or contribution.

Introduction

Explain the background and motivation of the project.

Checklist:

- ☐ What is the problem?
- ☐ Why is this problem important?
- ☐ Who may benefit from solving it?
- ☐ What is the goal of your project?

Related Work/Literature Review

Briefly describe existing studies, systems, datasets, or methods related to your project.

Checklist:

- ☐ Review at least 2–3 related sources.
- ☐ Explain how they are connected to your project.
- ☐ Mention what your project does similarly or differently.
- ☐ Include citations or references.

Problem statement

Clearly define the task your project solves.

Checklist:

- ☐ State the input.
- ☐ State the expected output.
- ☐ Define the AI/ML task type.

Task type:

- ☐ Classification
- ☐ Regression
- ☐ Clustering
- ☐ Recommendation System
- ☐ Computer Vision
- ☐ Natural Language Processing
- ☐ Time Series Analysis
- ☐ Generative AI
- ☐ Other: _____

Example format:

Given ABC, our system aims to predict / classify / generate / recommend EFG.

Dataset

Describe the data used in the project.

Checklist:

- ☐ *Name or describe the dataset.*
- ☐ *State the data source.*
- ☐ *State the number of samples, if known.*
- ☐ *Describe the main features/columns.*
- ☐ *Describe the target variable, if applicable.*
- ☐ *Mention any data limitations.*

Data source:

- ☐ *Public dataset*
- ☐ *Collected manually*
- ☐ *Web scraping / API*
- ☐ *Synthetic / generated data*
- ☐ *Other: _____*

Methodology

Explain how the project was implemented.

Checklist:

- ☐ *Describe preprocessing steps.*
- ☐ *Describe feature engineering, if any.*
- ☐ *Describe the model(s) or algorithm(s) used.*
- ☐ *Explain why the method is suitable.*
- ☐ *Describe the train/test split or validation method.*
- ☐ *Mention tools and libraries used.*

Experiments and Results

Present the results of your model or system.

Checklist:

- ☐ State the evaluation metric(s).
- ☐ Show the main results in a table, chart, or figure.
- ☐ Compare at least one baseline or simple method, if possible.
- ☐ Explain what the results mean.

Possible metrics:

- ☐ Accuracy
- ☐ Precision / Recall / F1-score
- ☐ Mean Squared Error / Mean Absolute Error
- ☐ R² Score
- ☐ Confusion Matrix
- ☐ Clustering score
- ☐ Human evaluation
- ☐ Other: _____

Discussion

Interpret the results and reflect on the project.

Checklist:

- ☐ What worked well?
- ☐ What did not work well?
- ☐ Why do you think the model/system performed this way?
- ☐ What are the limitations?
- ☐ What could be improved?

Output/Final output

Describe what your project produced.

Checklist:

- ☐ Report
- ☐ Source code
- ☐ Trained model, if applicable
- ☐ Dashboard / visualization, if applicable
- ☐ Web / mobile prototype, if applicable

- ☐ Chatbot / assistant, if applicable
- ☐ Demo screenshots, if applicable

Conclusion

Summarize the project in one short section.

Checklist:

- ☐ Restate the problem.
- ☐ Summarize the method.
- ☐ Summarize the main findings.
- ☐ Mention future work.

References

List all sources used in the project.

Checklist:

- ☐ Papers
- ☐ Articles
- ☐ Dataset links
- ☐ Documentation
- ☐ GitHub repositories, if used

Use a consistent citation format.

Appendix (optional)

Include extra materials/explanation/etc. if needed.

Checklist:

- ☐ Extra figures
- ☐ Extra tables
- ☐ Model parameters
- ☐ Screenshots
- ☐ Additional experiments
- ☐ Link to code repository

